



UNIVERSITY OF
CENTRAL FLORIDA

AST 5765C - Advanced Astronomical Data Analysis

Section: 0001

College of Sciences
Department of Physics

Course Information

Term: Fall 2026

Class Meeting Days: MWF

Class Meeting Time: 02:30PM - 04:20PM

Class Meeting Location: HEC 0110

Modality: P

Credit Hours: 3.00

Instructor Information

Name: Joseph Harrington

Title: Pegasus Professor

Office Location: PSB 441

Office Hours

By arrangement over summer

Phone: 407-823-3416

Email: jh@ucf.edu

Communication Expectations

Use WebCourses messaging to reach course staff. This increases the speed of responses over direct email, helps us keep all messages in one place, and protects your privacy. In an emergency, use Teams messaging to the TA or instructor, pointing to your WebCourses message. Until our WebCourse is activated, email the instructor.

Course Description

AST 5765C COS-PHYSICS 3(3,1) Advanced Astronomical Data Analysis: PR: MAC 2313, a 3000-level or higher course in astronomy or planetary science, ability to write simple computer programs, or C.I. Advanced astronomical data formation and acquisition, detector physics, measurement extraction, error analysis, modeling, computer programming, statistics, interpretation, and written and oral presentation of results. Fall.

Split-Level Course

This is the graduate version of this class. Undergraduates taking it may drop down to the undergraduate version at any time via an approved petition. Grades will be transferred, counting graduate-only work as extra credit.

Formal prerequisites

MAC 2313 (calculus)

Ability to write simple computer programs (see below)

An upper-level course in astronomy or planetary science, or CI

Computer Required in Class

Nearly all classwork is done on your computer, including in-class demonstrations. **You must bring a laptop to every class. We will be coding in class every day.**

Required:

- Not a tablet or phone, a real computer with a keyboard, mouse/trackpad
- Linux, MacOS, or Windows (in that order of preference, if you have options)
- Screen big enough for you to work in several windows next to each other
 - If you have options, a 14" 1080p or better screen helps
- For certain things, an external mouse or trackball and 3 buttons help
- At least 20 GB free storage
- 8+ GB RAM
- Able to support virtualization in 64 bits
- Fast enough to run course software (most machines are)

- Power sufficient for 2 hours

We will provide instructions to download and install the course computing environment. You must be able to install and configure software, following basic instructions. Training to work in the provided environment occupies the first two weeks of the course.

Coding Requirement and Assessment

We use the Python language. Students will need to become functional in Python within the first two weeks of the course, so **programming proficiency is required** for success in this course. Students who have taken the course without programming experience have struggled a great deal. Some have failed. **An on-paper, no-aids programming evaluation will be given on the first day of class.** It does not matter what language or packages you know for this assessment. **Students not demonstrating proficiency will be disenrolled.**

Preclass Coding Preparation

If you are in doubt about your coding ability, contact the instructor now for resources to prepare or to take a sample assessment to determine whether you are ready. You will still need to pass the official assessment given on the first day.

If you want to get ahead, learn Python and the basic use of its numpy, scipy, and matplotlib packages. Assignments and lectures from PHZ 3150 are available from the instructor to help guide your preparation. Again, this is not necessary if you know how to code, it will just make the first few weeks of the course easier.

If you are not ready for this course and want to take a course to prepare, the best course to take is PHZ 3150 Introduction to Numerical Computing. That course was designed specifically to prepare for this course. Conversely, if you took PHZ 3150 and got an A or a B, and you remember it, you are well prepared for this course.

Student Learning Outcomes

Those who successfully complete this course will be able to:

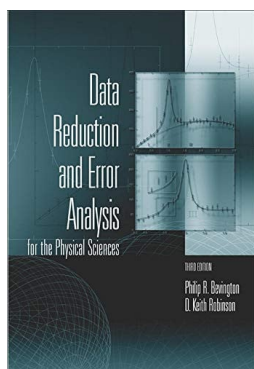
1. Understand basic statistics and error analysis as used in the physical

sciences

2. Extract physical measurements and error estimates from raw data
3. Find, educate themselves about, and select appropriate numerical analysis methods
4. Fit a theoretical model to the measurements, including uncertainties and signal-to-noise ratios
5. Draw scientifically-valid conclusions from the measurements
6. Manage and carry out online work with large amounts of data
7. Present scientific results

Subject to change until course begins. The course is being updated this summer to include AI coding at some level.

Required Course Materials and Resources



Data Reduction and Error Analysis for the Physical Sciences

ISBN: 9780072472271

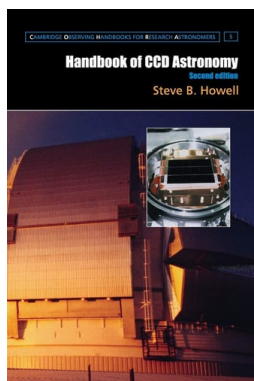
Authors: Bevington, Philip, Robinson, D. Keith

Publisher: McGraw-Hill Education

Publication Date: 2002-07-23

Edition: 3

Notes: THE standard reference on error analysis. Out of print. Available used online for \$30-50, and often cheaper. Expensive to buy new.



Handbook of CCD Astronomy

ISBN: 9780521617628

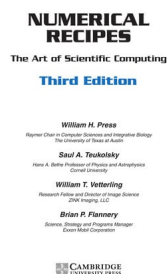
Authors: Steve B. Howell

Publisher: Cambridge University Press

Publication Date: 2006-03-02

Edition: 2

Recommended Course Materials



Numerical Recipes 3rd Edition The Art of Scientific Computing

ISBN: 9780521880688

Authors: William H. Press

Publisher: Cambridge University Press

Publication Date: 2007-09-06

Edition: 3

No need to buy (but it's a great reference to own).
Publisher's website has full text for free. We will not use code from the book.

Online Access: <https://numerical.recipes/book.html>

Class Schedule

Approach

We cover the following topics in roughly this order:

1. Computers, programming, online management
2. Introductory statistics and modeling
3. Array detectors and corrections, image analysis
4. Photometry
5. Spectroscopy
6. Fitting
7. Project

Data are information, and computers are information manipulators. You should therefore embrace the power of computation and become as expert in computing as in your chosen field. This class approaches the computer as a tool to master and command rather than a black box to fear. This topic is continual throughout the course.

In addition, throughout the course, we will read about and discuss a series of advanced topics in computational data analysis.

Course Schedule

This schedule is approximate. We may get ahead or fall behind.

Mondays and Wednesday are lecture days. Fridays are practicum, in-class demos, and help. To account for holidays, two Fridays late in the semester are lecture days. If the university shuts down (e.g., due to weather), additional Fridays may be converted into lecture days. Attendance is required on all scheduled class days, including Fridays.

THIS SCHEDULE WILL LIKELY CHANGE! In particular, which Fridays are lecture days is not finalized, and the project due date is not finalized. The project due date may shift to the beginning of finals week to allow more time for grading. This will depend on how well we can use AI to assist with project evaluation, and how much time that saves us, which we will not know until we have some experience with homework evaluation using AI.

Course Schedule

Le ct.	Date	M W F	Topic	Reading	Assignment
Tools and Theory					
1	24 Aug	M	Introduction, Software Install	Handout	
2	26 Aug	W	Linux	Handout	1 (Unix) pre-hw
3	31 Aug	M	First Peer Evaluation	SciPy docs	1 (Unix)
4	2 Sep	W	Functions, Coding Style, Measurements, Errors	Matplotlib tut., Pydatatut 1-2, readings on handout	
Photometry					
5	9 Sep	W	Measurement, Prob., and Stats	Bev. Ch 1,2	2 (SciPy)
6	14 Sep	M	Prob. Distributions, Error Anal	Handout, Pydatatut 3-4	3 (programming)
7	16 Sep	W	Fitting	Bev. Ch 3,4	4 (stats)
8	21 Sep	M	Fitting and Astronomical Meas.		
9	23 Sep	W	CCDs	Bev. Ch 6	5 (fitting)
10	28 Sep	M	Infrared Arrays, Sky, and Corrections	How. Ch 1,2, Press 14.1-14.3, 14.8, Pan-Starrs	
11	30 Sep	W	Combining Frames	Press 15.6, 15.8	6 (S/N, 2DGauss)
12	5 Oct	M	Flat Fields	Press 15.6, 15.8	

Le ct.	Date	M W F	Topic	Reading	Assignment
13	7 Oct	W	Coding an Analysis		7 (setup, dark)
14	12 Oct	M	Convolution, Point-Spread Func.	Press 15.6, 15.8	
15	14 Oct	W	Project, PSF, Centering	Press 15.6, 15.8	8 (sky)
16	19 Oct	M	Aperture Photometry	How. Ap C	
17	21 Oct	W	Astronomical Signal-to-Noise Calcs		9 (flat field)
18	26 Oct	M	Presenting Data		
Spectroscopy					
19	28 Oct	W	Presenting Research: Paper Writing		12 (publishing)
20	2 Nov	M	Bad Pixels		
21	4 Nov	W	Coding, Spectroscopy Introduction		10 (centering)
22	9 Nov	M	Spectroscopy	Horne paper	
23	13 Nov	F	Gratings and Systematics		11 (photometry)
24	16 Nov	M	Interpolation and Correlation		
25	18 Nov	W	Wavelength Calibration		(Project)
Project					
26	20 Nov	F	Bayesian Stats and Markov Chains		(Project)
27	30 Nov	M	Parameter Error Estimation		
28	2 Dec	W	Project Help, Postmortem		
	4 Dec	F	Project help session (time TBD)	(Optional)	(Project)
	8 Dec	T	Project help session (time TBD)	(Optional)	(Project)
	9 Dec	W	Project discussion 1 - 3:50 pm		Project Due 10a

Holidays

Date(s)	Day	Event
7 Sep	M	Labor Day
11 Nov	W	Veterans Day

Date(s)	Day	Event
23-27 Nov	M-F	Thanksgiving / Fall Break

Course Assessment and Grading Procedure

We will use peer evaluation (not grading) of homework assignments. Your participation in the peer evaluation counts toward your grade.

There is a major final project. We are evaluating using in-class exams.

This information will change!

Evaluation Weighting

Evaluation	Weight
Homework	45%
Attendance	5%
Discussion participation	5%
Peer evaluation participation	10%
Quizzes	10%
Project results	13%
Project execution (code quality)	5%
Project paper	7%

Grading Scale

Grading Scale

Grade	-			Straight			+		
A	90	-	93 1/3	93 1/3	-	100			
B	75	-	79.99...	80	-	84.999...	85	-	89.99...
C	60	-	64.99...	65	-	69.999...	70	-	74.99...
D	50	-	53 1/3	53 1/3	-	56 2/3	56 2/3	-	59.99...
F	below 50%								

Grading Scale

Letter Grade	Percentage
A	94-100%
A-	90-93%
B+	87-89%
B	84-86%
B-	80-83%
C+	77-79%
C	74-76%
C-	70-73%
D+	67-69%
D	64-66%
D-	61-63%
F	0-60%

I used the top scale in the past and I am assessing whether course changes will change that to the bottom scale.

To encourage group participation, grades will not be curved. It is possible for everyone to get an A. It is also possible for everyone to fail (but I hope not!). All reasonable questions regarding grading are welcome.

Group Peer Evaluation

Data analysis coding is among the most difficult to do correctly, because there are myriad “moving parts” to track, and because wrong code often does not look much different from correct code. Learning what to do from lists presented in lecture is ineffective; we learn coding experientially. We learn nearly as fast from the experience of others (both positive and negative) as from our own, and with much less effort (we didn’t have to write that code!).

Therefore, we use open peer evaluation as a primary learning tool. About once in the semester, each student’s homework will be displayed on screen, and we will discuss, with the solutions and grading rubric in hand, how the assignment was coded, and often multiple other ways it could have been coded. Allowing your colleagues to learn from your insights and mistakes will let you learn from theirs, leading quickly to better results for everyone. To date we have never had

an incident of disrespect during the peer evaluation, and we expect (and require) this to continue.

The student assigned to evaluate your code will use Zoom to share their screen for projection.

Simultaneously, each student will evaluate another's homework using the grading rubric. Going through the rubric, solutions, and another's code in detail leads to a much deeper understanding and a longer retention of the material. Participation in the peer evaluation is graded. The final homework grades are given by the course staff and frequently differ from the peer evaluation rubrics.

Homework Format

Hand in all homework on WebCourses as a single tar.gz file per assignment, unless otherwise stated, including prose and math. To make the peer evaluation and grading easier and to promote good coding habits, there are strictly enforced homework format and style rules. See the handout on that topic.

The comments in homework feedback are often about coding style or homework format, so a close reading is important to avoid losing points for the same errors on multiple assignments.

Working Effectively

There will be approximately weekly homework assignments and project work. It is critical that you do the homework and readings by the beginning of class on the due date, as we discuss answers in class. Your personal understanding is what counts in the discussions, and discussions count toward your final grade. Since answers will be discussed in class, **no late homework will be accepted.**

The **homework solutions** are a critical teaching element. **You must read them**, including all comments in code, each week. They contain discussion of how and why to do important things, knowledge you will need for later assignments.

Not all information needed to do the assignments is covered in lecture or the texts. You must find sources for and read about programming and numerical

methods on your own. The Recommended Textbook is a good place to start, as is the web.

Compared to most physics courses, this course is heavy on skills, methods, and experience. These are taught with practice on real data in the homework assignments. Unlike the “example” datasets of other courses, real data has real problems with it that you will have to overcome. You should budget significant time each week to work on your homework and project. Assignments will depend heavily on prior work done in the class, so skipping work is not very useful: you'll be doing the work anyway in order to do later assignments, so it makes sense to do it in time to get credit for it.

Remember that debugging can take a long time, so start your assignments early! While time spent on the class varies a great deal according to students' prior programming experience, you should expect to spend an average of 6-10 hours per week outside of class on this course, and more if this is your first exposure to computing “for real”.

Project

In October, each student will start a final project based on real data. You will apply the methods learned in the course to produce a measurement and reach a scientific conclusion. Three components of the project together contribute 30% of your final grade: a paper, which will follow the format of *The Astrophysical Journal*, what your coded analysis routines produce (results), and how well your analysis routines are coded and documented (execution).

Video Recording

We video-record each class. This allows students to make up missed classes and to review material. The videos may be posted for the class without redaction. Any videos posted publicly will first be scrubbed of material that shows student faces or grades. Students may not sit in the camera's field of view.

Policies for Course Grade

Attendance

Class attendance - both lecture and practicum days - is mandatory. Preparation for and participation in class discussion and activities counts toward the final grade. You must be an active learner by doing the demos and exercises along with the instructor on your computer; don't just watch.

Those unable to attend class in person on a particular day may, by arrangement with the instructor, attend via the internet. This is intended to avoid spreading disease and to enable attendance during official travel, and should not be used when it is physically reasonable to attend class. **If you have symptoms but you can function reasonably well, please attend remotely.**

Missed/Late Assignments

The homework assignments are due at the beginning of class on the due date. **No late work will be accepted**, because we discuss the assignments in class, so **PLAN AHEAD**. If an excused absence is known sufficiently in advance, homework must be handed in by the deadline. If it is unexpected, the grade will be dropped, or the portion completed will be graded out of that portion's points. In many cases, the homework must still be done if the subsequent homework depends on it, as is frequently the case.

Reading should be done **before** the class indicated. There will be reading quizzes to motivate reading. These can be made up, for excused absences.

Makeup Work Policy

Making up missed work will only be permitted for University-sanctioned activities, legally required absences (jury duty, court dates, military service, etc.), and serious medical or family reasons. Justifying documentation may be required (in advance for University-sanctioned activities).

At the discretion of the instructor, the make-up may take any reasonable and appropriate form including, but not limited to allowing a late submission, a similar replacement item, or dropping the grade.

The grading scheme drops the lowest homework and quiz grades. This allows you not to hand in one homework and not to take one quiz if you wish to have an unexcused absence, without averaging a 0 into your grade. We strongly urge you not to take this option, but rather to use it to improve your grade by dropping your worst scores.

Academic Honesty, Sharing, and Information Sources

We will follow the letter and spirit of the UCF Golden Rule. Research in astronomy and physics relies on taking advantage of resources developed elsewhere: software libraries, descriptions of methods, etc. Unless we state otherwise, please use such external sources in your work. However, there are several conditions:

1. All math, code, and text answers must be your original work. You may (and should) discuss the relevant general topics with each other, but you may neither give nor receive specific help on, nor share, assigned work. Sharing code, even to the extent of making it visible on-screen or reading it from someone else's screen, is not allowed, regardless of whether the other party gives permission. This includes help from others not in the class.
2. For coding problems, the portion of the answer relevant to the problem must be your original work. For example, if the question asks you to subtract two images, you must write the code to do the subtraction but you may use third-party code to read the images from files.
3. You must have legal permission to use an external source (assumed if publicly posted).
4. You **MUST** give credit to all external sources on a problem-by-problem basis. Credits must include the name of the item, a sentence fragment describing it if it is not obvious from the name, its author(s), year of authorship, and location (e.g., the name, volume, and pages of a journal article, or the URL of a software package distributed online).
5. As with any scientific research project, you alone are responsible for the output: if you download a package that claims to do something and it has a bug that gives the wrong answer, the answer is wrong and you will be marked accordingly.
6. Work you did prior to the start of the course may not be handed in for grade (talk to the instructor for exceptions).

In addition, see the required statements on academic integrity, below.

Artificial Intelligence (AI) Use Policy

Use of Generative Artificial Intelligence (GenAI) is governed on an assignment-by-assignment basis. For each assignment, you must carefully review the instructions to determine whether AI-generated content is permitted. In some cases, you may be permitted (though not required) to use GenAI. When using GenAI, you must disclose specifically where and how it was used in the process of completing the assignment. Unless explicitly allowed, all components of your work must be fully prepared by you. It should also be cited. For example, text generated from Copilot should include a citation such as: "Copilot. Accessed 2023-12-03. Prompt: 'Summarize the Geneva Convention in 50 words.' Generated using Copilot." Any attempt to represent GenAI output inappropriately as your own work will be treated as plagiarism. Ultimately, students are responsible for the accuracy and veracity of submitted work whether AI-generated or not.

Make-up Assignments for Authorized University Event or Co-curricular Activities

Students who represent the university at an authorized event or activity (for example, student athletes) and who are unable to meet a course deadline due to a conflict with that event must provide the instructor with documentation in advance to arrange a make-up. No penalty will be applied if the student gives advance notice and communicates with the instructor following UCF policy. In the case of an authorized university activity, it is your responsibility to show the instructor a signed copy of the Program Verification Form for which you will be absent, prior to the class in which the absence occurs or due date you need extended. For more information, see [UCF Policy 4-401](#).

Make-up Assignments for Religious Observances

A student who desires to observe a religious holy day of his or her religious faith must notify their instructor as soon as practicable prior to the observance. Students who are absent because of religious observances will be permitted a reasonable amount of time to complete any missed work. For more information, see [UCF Regulation 5-020](#).

Disability Access & Accommodations

The University of Central Florida is committed to providing equal access to all students with disabilities (ADHD, learning disabilities, Autism, chronic medical conditions, physical disabilities, etc.). To receive consideration for reasonable disability-related course accommodations, disabled students must contact Student Accessibility Services (SAS) and complete the steps required for SAS to review accommodation requests. More information can be found on the UCF [Student Accessibility Services](#) website under the Start Here tab or by contacting SAS directly (Ferrell Commons 185; sas@ucf.edu; Phone - 407-823-2371).

Approved accommodations are shared with course instructors via the SAS Course Accessibility Letter. Implementing certain accommodations may require discussion about specific considerations of the course design, course learning objectives, and the individual academic and course challenges experienced by the student. While students with disabilities or chronic health needs are also encouraged to discuss any course concerns with professors in addition to contacting SAS, professors are not required to facilitate disability-related adjustments to the course unless the professor has received a Course Accessibility Letter from SAS that outlines approved accommodations.

Academic Integrity

Students should familiarize themselves with UCF's Code of Conduct at Student Conduct and Integrity Office. According to Section 1, "Academic Misconduct," students are prohibited from engaging in:

1. Academic misconduct is defined as any submitted work or behavior that obstructs the instructor of record's ability to accurately assess the student's understanding or completion of course materials or degree requirements (e.g., assignment, quiz, and/or exam). Examples of academic misconduct include but are not limited to: plagiarism, unauthorized assistance to complete an academic exercise; unauthorized communication with others during an examination, course assessment, or project; falsifying or misrepresenting academic work; providing misleading information to create a personal advantage to complete course/degree requirements; or multiple submission(s) of academic work without permission of the instructor of record.

2. Any student who knowingly helps another violate academic behavior standards is also in violation of the standards.
3. Commercial Use of Academic Material. Selling of course material to another person and/or uploading course material to a third-party vendor without authorization or without the express permission of the University and the instructor of record. Course materials include but are not limited to class notes, the instructor of record's slide deck, tests, quizzes, labs, instruction sheets, homework, study guides, and handouts.
4. Soliciting assistance with academic coursework and/or degree requirements. The solicitation of assistance with an assignment, lab, quiz, test, paper, etc., without authorization of the instructor of record or designee is prohibited. This includes but is not limited to asking for answers to a quiz, trading answers, or offering to pay another to complete an assignment. It is considered Academic Misconduct to solicit assistance with academic coursework and/or degree requirements, even if the solicitation did not yield actual assistance (for example, if there was no response to the solicitation).

Responses to Academic Dishonesty, Plagiarism, or Cheating

Students should also familiarize themselves with the procedures for academic misconduct in UCF's student handbook, [The Golden Rule](#). UCF faculty members have a responsibility for students' education and the value of a UCF degree, and so seek to prevent unethical behavior and respond to academic misconduct when necessary. Penalties for violating rules, policies, and instructions within this course can range from a zero on the exercise to an "F" letter grade in the course. In addition, an Academic Misconduct report could be filed with the Office of Student Conduct and Academic Integrity, which could lead to disciplinary warning, disciplinary probation, or deferred suspension or separation from the University through suspension, dismissal, or expulsion with the addition of a "Z" designated on one's transcript.

Being found in violation of academic conduct standards could result in a student having to disclose such behavior on a graduate school application, being removed from a leadership position within a student organization, the recipient of scholarships, participation in University activities such as study abroad, internships, etc.

Let's avoid all of this by demonstrating values of honesty, trust, and integrity. No grade is worth compromising your integrity and moving your moral compass. Stay true to doing the right thing: take the zero, not a shortcut.

Title IX

Title IX prohibits sex discrimination, including sexual misconduct, sexual violence, sexual harassment, and retaliation. If you or someone you know has been harassed or assaulted, you can find resources available to support the victim, including confidential resources and information concerning reporting options at [Let's Be Clear](#) and [UCF Cares](#).

For more information on access and community engagement, Title IX, accessibility, or UCF's complaint processes contact:

- Title IX – ONAC – [Office of Nondiscrimination & Accommodations Compliance](#) & askanadvocate@ucf.edu
- Disability Accommodation – Student Accessibility Services – [Student Accessibility Services](#) & sas@ucf.edu
- [Access and Community Engagement](#) (including the Ginsberg Center for Inclusion and Community Engagement, Military and Veteran Student Success, and HSI Initiatives)
- UCF Compliance and Ethics Office – [Compliance, Ethics, and Risk Office](#) & complianceandethics@ucf.edu
- The [Ombuds Office](#) is a safe place to discuss concerns.

Reporting an Incident or Issue

If you believe you have experienced discrimination by any faculty or staff member, contact the Office of Nondiscrimination & Accommodations Compliance via the [ONAC website](#) or at 407-823-1336. You can also choose to report using the UCF Integrity Line either anonymously or as yourself at 1-855-877-6049 or by using the [online form](#). UCF cares about you and takes every report seriously. For more information see the [Reporting an Incident or Issue Webpage](#).

Deployed Active-Duty Military Students

Students who are deployed active-duty military and/or National Guard personnel and require accommodation should contact their instructors as soon as possible after the semester begins and/or after they receive notification of deployment to make arrangements.

Campus Safety

At UCF's Public Safety and Police, safety is the top priority. Emergencies on campus are rare, but if one should arise, it's important to be familiar with some basic safety and security concepts.

- In an emergency, always dial 911.
- Every UCF Classroom has an Emergency Procedure Guide posted on a wall near the door, which will show you how to respond to a variety of situations. This guide can also be found online [here](#).
- In the event of an active threat, remember **AVOID, DENY, DEFEND**. Choose the best course of action and act immediately. Watch the video [here](#) to learn more.
 - **AVOID**. Pay attention to your surroundings and have an exit plan. Get as much distance and as many barriers between you and the threat as quickly as possible.
 - **DENY**. When avoiding is difficult or impossible, deny the threat access to you and your space. Lockdown by creating barriers, turning the lights off and remaining quiet and out of sight. Make sure your phone is silenced, but do not turn it off.
 - **DEFEND**. When you are unable to put distance between yourself and the threat, be prepared to protect yourself. Commit to your actions, be aggressive and do not fight fairly. Do whatever it takes to survive.
- For emergencies on campus, UCF will utilize the [UCF Alert](#) system. All UCF students, faculty, and staff are automatically enrolled to receive these email and text alerts, however, it's a good idea to frequently ensure your [contact information is up to date](#).

Financial Aid Accountability

All instructors are required to document students' academic activity at the beginning of each course. In order to document that you began this course, please complete this activity by the end of the first week of classes or as soon as possible after adding the course. Failure to do so may result in a delay in the disbursement of your financial aid.